

create_placement

NAME

`create_placement`
Performs coarse placement on the current design.

SYNTAX

```
status create_placement
    [-floorplan]
    [-host_options host_option_name]
    [-use_seed_locs]
    [-effort very_low | low | medium | high]
    [-timing_driven]
    [-buffering_aware_timing_driven]
    [-congestion]
    [-congestion_effort low | medium | high]
    [-incremental]
    [-from combined_placement | macro_legalization | channel_sizing]
```

Data Types

`host_option_name` string

ARGUMENTS

- floorplan
Specifies that floorplanning placement should be used. This includes macro placement and special effort levels which are tuned for speed. There are a number of special application option controls for floorplanning. To get a list of them use `report_app_options plan.place.*` and `report_app_options plan.macro.*`.
One application option which you must be aware of is `plan.place.auto_create_blockages`. This controls whether soft, hard, or partial blockages are automatically created in thin channels between hard macros and block boundary. This is useful for congestion reduction. See man page of this application option for more details.
The macro placement style is controlled using the application option `plan.macro.style`. See the man page of this application option for more information.
- host_options host_option_name
Specifies that the given host option should be used for distributed processing. Host options are set with the `set_host_options` command. Note that not all operations performed by the `create_placement` command support distributed processing. The effect of this option depends on the design and the other option settings.
- use_seed_locs
Specifies that the current placement should be used as a seed for placement.
- effort very_low | low | medium | high
Specifies the CPU effort level for coarse placement. The default effort level is high.
The exception is that for on-edge and island style floorplanning placement, the default effort level is medium.
- timing_driven
Enables timing-driven placement. When used with the `-floorplan` option, you can adjust the behavior of the command with the `plan.place.timing_driven_mode` application option.
- buffering_aware_timing_driven
Enables timing-driven placement using an approximate timing model that estimates the effects of buffering long nets and high-fanout nets later in the flow. The netlist is not changed. Intended for use as an initial placement step that gives a better starting point for later timing optimizations.
- congestion
Enables congestion-driven placement mode.
When used with the `-floorplan` option, you can adjust the behavior of the command with the `plan.place.congestion_driven_mode` application option.
- congestion_effort low | medium | high

Specifies the effort level for congestion mode. The default effort level is medium. Expect a significant increase in runtime for high effort. This option can only be used with the -congestion option.

-incremental

Performs incremental placement on the design. Incremental placement tries to keep cells close to their current location. The -incremental option automatically enables the -use_seed_locs option. By default, the command can move cells far from their current location to improve placement QoR.

When -incremental is used in combination with -floorplan, the macro placement is legalized while trying minimizing the macro movements. Also, the standard cell placement is updated in incremental fashion. When also combined with the -congestion option, the command expects that the current placement is legal and that an up-to-date congestion map exists. This congestion map is used to increase the channel sizes between the macros in congested areas. Again, the standard cell placement is updated in an incremental fashion. See also the man page for the plan.place.congestion_driven_mode application option. For incremental freeform or hybrid macro placement, see also the -from option below.

-from combined_placement | macro_legalization | channel_sizing

Specifies the incremental level for incremental freeform or hybrid macro placement.

This option can only be used in combination with -floorplan -incremental and if the plan.macro.style application option is set to freeform or hybrid.

The value passed with this option indicates the stage to start the macro placement flow.

The hybrid macro placement normally starts with the placement of the on-edge macros. This step is skipped for incremental placement. The on-edge macro placement is preserved and only changes due to congestion driven channel sizing.

The freeform macro placement consists of multiple stages:

combined_placement

The macros and standard cells are placed together using the coarse placement engine. When -congestion is used, this is congestion driven.

macro_legalization

The macros are legalized, which makes sure that all macro placement constraints are satisfied. Also, the standard cells are placed incrementally, to adjust to the legalized macro locations.

channel_sizing

Congestion driven macro channel sizing, which increases channel sizes between macros where congestion is detected. This is followed by congestion driven standard cell placement. This stage is only run when the -congestion option is used.

The -from options indicates at which of these stages the incremental freeform placement starts.

So, the value combined_placement is the most aggressive and runs the full freeform flow while the channel_sizing entry point is the most subtle and only runs the last stage.

The default entry point is macro_legalization.

DESCRIPTION

The create_placement command performs coarse placement on the current design. With the -floorplan option, this includes the placement of hard macros as well as standard cells. To avoid placing hard macros, mark the hard macros as fixed with the set_attribute or set_fixed_objects command.

EXAMPLES

The following example runs the create_placement command in timing-driven mode:

```
prompt> create_placement -effort high -timing_driven
```

SEE ALSO

[check_host_options\(2\)](#)

[legalize_placement\(2\)](#)

[read_def\(2\)](#)

[report_app_options\(2\)](#)

[set_attribute\(2\)](#)

[set_host_options\(2\)](#)

[write_def\(2\)](#)

[place.coarse.*\(3\)](#)

[plan.place.congestion_driven_mode\(3\)](#)
[plan.place.auto_create_blockages\(3\)](#)
plan.place.*(3)
plan.macro.*(3)

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